

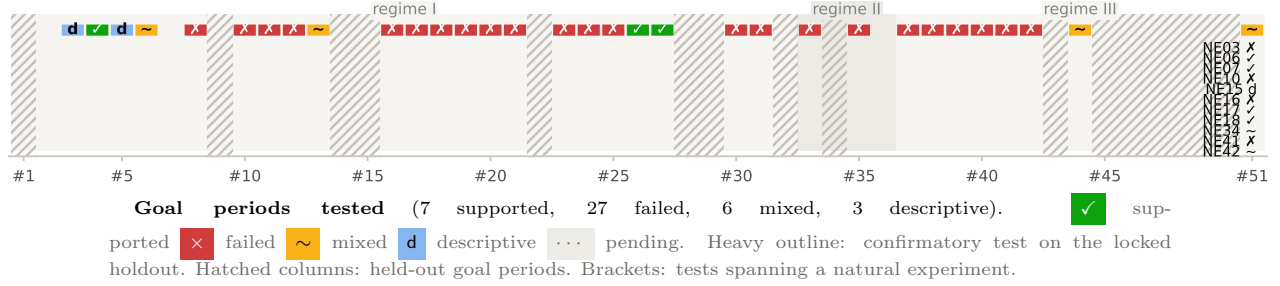
H39: Catalysts vs. fields

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

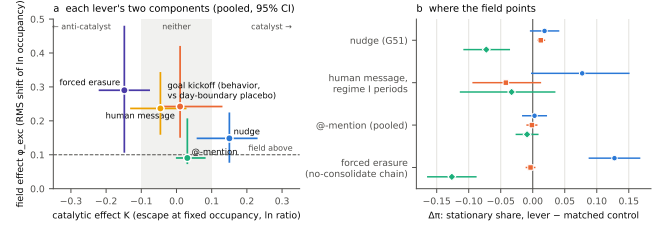
Bottom line. No operator lever is a pure catalyst: nudges both unstick and tilt agents, while human messages, mentions, erasures and goals act mainly as fields, on activity or on content depending on the regime.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Each operator lever was split into a *field* part (it shifts which behavior states agents occupy) and a *catalytic* part (it changes escape rates at fixed occupancy). Point levers used matched windows in 32 non-holdout periods; steps (26 kick-offs, 10 scaffold changes, room changes) used day-boundary placebos. A synthetic check came first: catalysts are recoverable from ~30 kicks, fields only from hundreds. No lever is a pure catalyst. Nudges are *both*: idle escape $\times 1.4$ –1.6, the stationary idle share falls 0.07 in G51, and catalytic fraction $\rho = 0.51$. That rejects HH52’s “catalyst”. Human messages are fields: chat→work in regime I, nothing in G51. @-mentions move content toward the message (+0.32 SD) but not behavior. Forced context erasure is an anti-catalytic field toward work (idle down in 8/8 periods). Kickoffs tilt content only in regime III. Scaffold steps: mostly nothing. Holdout not run.



Observables. (a) Pooled field strength ϕ_{exc} (placebo-corrected) against catalytic effect K , with 95% CIs. Dashed line and grey band: the 0.10 thresholds. Only the nudge reaches the “both” corner; human messages and erasure are fields, and erasure is anti-catalytic. (b) Where each field points: stationary share changes for work, chat and idle.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Partly useful. An operator gets a concrete rule. To unstick an idle agent, send one directed nudge: escape rises ~50%, and the agent also stays out of idle while it acts. To steer what an agent writes without changing how busy it is, mention it. To change topics, set a goal; that works in the large regime-III village. The physics label is softer than it sounds: the field/catalyst split depends on a convention, the nudge verdict rests on one period (G51), and the effects are individual and diluted. Switching the nudger off (08-21, post hoc) barely moved the swarm. “Messages always carry a field” is the robust, slightly deflating lesson. Prompt tweaks, room changes and most fixes showed nothing at day scale. Unconfirmed on the holdout.

PHYSICAL MODEL

Models 02/10: a kinetic Potts (Markov-jump) agent over states $s \in \{\text{work, chat, idle, consolidate}\}$ with Arrhenius rates k_{ij} , well depths G_i and symmetric barriers B_{ij} . A field h tilts the wells; a catalyst c lowers the barriers. Measured per lever: kicked and control transition matrices T^K, T^C in matched 30-min windows, their stationary distributions π^K, π^C , the escape probability $e_s = 1 - T_{ss}$, the field index ϕ , the catalytic index K , and the catalytic fraction ρ .

$$k_{ij} = \nu e^{-(B_{ij}-G_i)/T}, \quad \pi_i \propto e^{-G_i/T}$$

$$s_{ij} = \frac{1}{2}(\delta \ln k_{ij} + \delta \ln k_{ji}) \quad (\text{catalytic part})$$

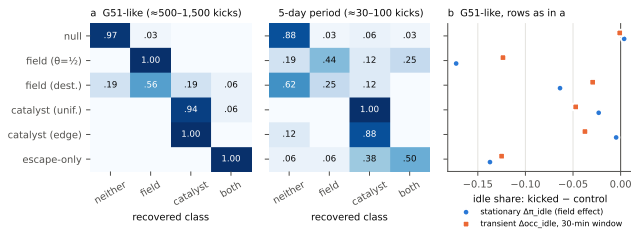
$$a_{ij} = \frac{1}{2}(\delta \ln k_{ij} - \delta \ln k_{ji}) \quad (\text{field part})$$

$$\phi = \text{SD}_{\pi^C} \left[\ln \pi_s^K - \ln \pi_s^C \right], \quad \rho = |K| / (|K| + \phi_{exc})$$

$$K = \ln \left(\sum_s \pi_s^C e_s^K / \sum_s \pi_s^C e_s^C \right)$$

RESULTS VS PREDICTIONS

- **P1 nudges = both** (mostly ✓): $K = +0.15$ [0.07, 0.23]; idle escape $\times 1.4$ (G51 $\times 1.57$); G51 $\Delta\pi_{\text{idle}} = -0.07$ [-0.11, -0.04]; $\rho = 0.51$. Pooled idle CI spans 0.
- **P2 human = both, toward chat** (×): a field (P 0.94) pointing chat→work in regime I; nothing in G51.
- **P3 mentions = field toward chat** (mixed): no consistent behavior direction; content drift toward the message +0.32 SD.
- **P4 erasure: browse up, type down** (×): idle -0.13, work +0.13, $K = -0.15$; idle down in 8/8 periods.
- **P5 kickoffs = content field** (3/4): 4/6 regime-III vs 0/13 regime-I kickoffs; behavior quiet.
- **P6 rooms** (inconclusive); **P7 scaffold** (mixed: 0/4 prompts, NE16 not catalytic, NE03 catalytic).
- **P9 nudger off 08-21** (post hoc, ✓): neither; idle escape -13%, occupancy flat.



Second figure. Synthetic validation at village sampling. (a) Share of replicates assigning each planted lever to each class: at G51 scale fields and catalysts are recovered (0.94–1.00); in a 5-day period catalysts still are, fields mostly are not. (b) The transient fallacy: a pure catalyst lowers the 30-min idle share without moving the stationary one.

GOAL PERIODS IN DETAIL

G03 d descriptive
 G04 ✓ supported
 G05 d descriptive
 G06 ~ mixed
 G08 × failed
 G10 × failed
 G11 × failed
 G12 × failed
 G13 ~ mixed
 G16 × failed
 G17 × failed
 G18 × failed
 G19 × failed
 G20 × failed
 G21 × failed
 G23 × failed
 +27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The nudge class rests on one period (G51; G38 has the same signs but is not significant). Fields are invisible below ~100 episodes, so “catalyst” or “neither” in small periods does not exclude a field. The field/catalyst split depends on a load-distribution convention; $|K| < 0.10$ is not interpretable. B4 states mean different things in regimes I and III. Mentions are partly common-cause. Design choices made after seeing data are flagged: the no-consolidate erasure chain, the content-TV check, and the post hoc P9 switch-off. 28 class-level tests: the nudge field alone does not survive Bonferroni.

ROUND 2

Run `confirm.py` after commit (expect low nudge power on #45–#50). Agent-centered content states for steps; a dose analysis of nudger re-firing; the 08-21 switch-off as a proper NE with day-matched controls.

No round-2 redirects yet.